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NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

First Semester M.Tech (VDE) (CBCS) Degree Examinations

December 2024 - January 2025

22VDE133 – SYSTEM ON CHIP (SOC) DESIGN

Duration: 3 Hours

Max. Marks:100

Note: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and One full question from Unit – III.

Unit – I		Marks	BT*	CO*	PO*
1. a)	Explain Moore's law under the following heads.				
	i) Statement				
	ii) Lives and Deaths of Moore's law				
	iii) In recent years				
	iv) Future of Moore's law	10	L*2	1	3
b)	Explain the advantages of System-on-Chip in detail.	10	L2	1	3
2. a)	With a neat diagram and table compare SOB, SOC & SOP technologies.	10	L2	1	3
b)	Write a note on Scaling. Show the result of technology scaling.	10	L2	2	3
3. a)	With a neat diagram, explain the typical SoC Architecture.	10	L2	2	3
b)	Draw the graphical representation for the three SoC IPs w.r.t to the following.				
	i. Reusability				
	ii. Portability				
	iii. Predictability				
	iv. Performance				
	v time to market	10	L3	2	3
Unit – II					
4. a)	How to reduce active and standby energy in case of energy-aware memory system design.	10	L3	3	3
b)	What are MPSoCs? Why MPSoCs are required? Outline the challenges associated with MPSoCs.	10	L2	3	3
5. a)	Explain software from the viewpoint of the application programmer, software architecture and design reuse viewpoint.	10	L2	4	3
b)	Illustrates the components of power consumption in a simple CMOS circuit.	10	L2	4	3
6. a)	With a neat diagram, explain the cache architecture of multi ported shared cache, private caches, and CCC.	10	L2	3	3
b)	What is quality-of-service (QoS)? Explain Hardware Architectures.	10	L2	4	3
Unit – III					
7. a)	With a neat diagram explain, HW/SW Co-design at System-Level.	10	L2	5	3
b)	Summarize different types of design validation and verification.	10	L2	5	3
8. a)	Explain Voice/Tone Processing Unit and Packet Processing Unit.	10	L2	5	3
b)	Illustrate VoIP SoC with a neat diagram.	10	L3	5	3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
